

Acceptance Testing UAT Execution & Report Submission

Date	29 June 2026
Team ID	
Project Name	OptiCrop : Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

1. Purpose of Document

The purpose of this document is to verify that the **OptiCrop – Smart Agricultural Production Optimization Engine** satisfies the functional and non-functional requirements specified by the end users. User Acceptance Testing (UAT) was conducted to validate the accuracy of crop recommendations, usability of the web interface, input validation, system reliability, and overall performance before project submission.

2. Defect Analysis

This report shows the number of resolved or closed bugs at each severity level, and how they were resolved

Resolution	Severity 1	Severity 2	Severity 3	Severity 4	Subtotal
By Design	0	0	0	1	1
Duplicate	0	0	0	0	0
External	0	0	0	0	0
Fixed	0	1	2	3	6
Not Reproduced	0	0	0	0	0
Skipped	0	0	0	0	0
Won't Fix	0	0	0	0	0
Totals	0	1	2	4	7

3. Test Case Analysis

This report shows the number of test cases that have passed, failed, and untested

Section	Total Cases	Not Tested	Fail	Pass
User Interface	5	0	0	5
Input Validation	7	0	0	7

Machine Learning Prediction	6	0	0	6
Flask Backend	4	0	0	4
Model Loading (pickle)	2	0	0	2
Crop Information Display	4	0	0	4
Performance Testing	3	0	0	3
Total	31	0	0	31

User Acceptance Testing was successfully completed for the OptiCrop system. All functional modules, including data input, validation, machine learning prediction, crop recommendation, and result display, performed as expected. The application generated accurate crop recommendations with an overall model accuracy of **99.32%**. No critical defects were identified during testing, and the system was accepted for project submission

